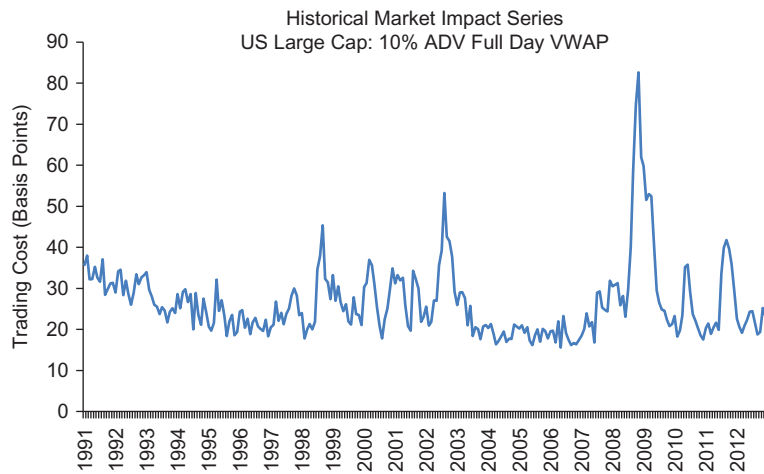




■ Figure 12.5 Historical Back-Testing.

spread size of 1/8th or 1/4 in the 1990s many now only incur a trading cost of a few cents and result in a profitable strategy.

These historical cost index series could be used in conjunction with a portfolio manager's current costs to determine an appropriate customized series via our back-propagation technique ([Equation 12.4](#)).

Market Impact Simulation

In this section we present a market impact simulation experiment to highlight the difficulty in developing stock specific market impact models. The approach also presents techniques that can be used by investors to test and evaluate different types of financial models, not just market impact models. The techniques are centered around simulating market conditions and prices following a defined model and model parameters. The exercise then sets out to estimate the model parameters based on the simulated data. If the process can uncover the "true" parameter values then the modeling and estimation approach is reasonable. Otherwise, even if the model is perfect, the difficulty in estimating model parameters may cause the approach to be unusable. Remember in this case that, since we are simulating data based on a specified model, the "true" parameters are those parameters that were used to simulate the data.